Optimal Sensor Placement for Wastewater Network Monitoring

Placing sensors usefully in a wastewater network depends on two distinct bodies of evidence: knowing where deterioration risk concentrates, and knowing where a sensor is actually informative once located. Six recent studies address these questions from complementary angles – condition-assessment technology, deterioration-factor identification, a quantified probabilistic condition model, low-data risk prioritisation, and two sensor/monitoring placement methods – and are synthesised below.

Condition-assessment technology and deterioration factors

Ma et al. (2026) review AI-driven sewer condition assessment and show that, while closed-circuit television (CCTV) remains the predominant inspection technology, its reliance on manual interpretation motivates automated approaches. Traditional machine-learning models built on structured asset data show a wide performance spread: logistic regression achieves only 66.6% accuracy on binary failure classification, while tree-based models perform substantially better (random forest ≈78.3%, XGBoost ≈92.9%), and a hybrid ANN–random-forest model exceeds 99%. Deep-learning models applied directly to CCTV imagery report strong defect-detection performance (YOLO-family mAP >85%; one segmentation model reaching 97% mean pixel accuracy), but the review stresses that class imbalance in training data – common defects such as cracks and deposits dominate over rarer, more consequential ones – limits how far these figures generalise. Across the model-input categories the review tabulates, physical attributes (age, diameter, length, material, slope) appear in nearly all studies, whereas hydraulic (<20%) and chemical (2 studies) attributes are rarely available, reflecting a broader data-cost asymmetry.

Malek Mohammadi et al. (2020) systematically review 23 sewer condition-prediction studies (2001–2019) and frame deterioration with a bathtub-curve hazard function: an elevated early-life failure rate from installation defects, a low and roughly constant useful-life rate, and a rising wear-out rate as pipes age. Age is the most consistently significant predictor across studies. Material interacts with environment rather than acting alone (concrete resists abrasion but not acid; clay resists acids; PVC/HDPE resist both; reinforced concrete is generally most resistant), and root intrusion increases with age, raising hydraulic roughness and reducing capacity. Diameter, pipe length, and burial depth are the three factors with genuinely contradictory findings across the literature: larger diameter is variously reported as protective (better resistance to ring/crushing stress) or confounded with burial depth; longer segments are usually higher-risk (more joints, more blockage potential); and depth effects are significant in some studies and absent in others. Combined sewer systems deteriorate faster than separated sanitary systems, and industrial land use accelerates deterioration relative to residential. The review's key methodological conclusion is that these contradictions are partly an artefact of data availability – physical attributes are cheap and almost universally recorded, while environmental and operational attributes are expensive to collect and largely absent from utility databases, so different studies are effectively built on different, incomplete subsets of the true factor set.

A quantified condition model

Ma et al. (2025) operationalise this factor catalogue for a specific network, building a Bayesian-network (BN) condition model from 23,000 integrated Hong Kong pipe records spanning 14 physical, environmental, and climate-related factors (15 nodes, 31 links, structure learned via a

Bayesian Information Criterion score). Mutual-information sensitivity analysis ranks pipe age as the dominant factor (MI = 0.124), followed by diameter (0.032), population density (0.027), and soil type (0.024). Deposits are the most common recorded defect (20.19%), followed by fracture (13.84%), displaced joint (13.39%), and crack (12.09%). The model's most operationally relevant result is a compounded-risk scenario: baseline critical-condition probability is 17%, rising to 72% when all four dominant factors are simultaneously at their highest-risk state (age > 50 years, diameter ≤ 200 mm, population ≤ 15,000, soil in fill or granitic rock) – demonstrating that a small set of jointly-considered factors can substantially sharpen risk targeting beyond any single-factor threshold.

Risk-based prioritisation under limited data

Lundberg et al. (2026) address the more common utility situation of incomplete condition data by proposing a two-tier multi-criteria decision analysis (MCDA) model requiring only attribute data and structured stakeholder input. A weighted-sum score ($W_i = IS_i/\Sigma IS_i$, applied to workshop-elicited sub-criterion values) combines a probability-of-failure (POF) score from four criteria – material, age, length, and diameter – with a consequence-of-failure (COF) score from eight proximity-based criteria (buildings, roads, railways, water bodies, water-protection areas), into a risk-of-failure grade via a 5×5 risk matrix. Tier I uses this to prioritise which pipes receive a CCTV inspection first; Tier II then folds the resulting structural grade back into an updated POF to prioritise rehabilitation, so municipalities with little or no existing CCTV coverage can still run the model from Tier I alone. Applied to a 571.3 km, 15,044-pipe network in Kungsbacka, Sweden, the model scored 97.3% of pipes and produced a working inspection and rehabilitation plan within six months – evidence that a defensible, explainable risk ranking is achievable without a calibrated hydraulic model or complete condition history.

Sensor placement and network observability

Simone et al. (2023) compare two strategies for locating a fixed number of water-quality sensors to maximise contaminant-detection reliability: a deterministic global optimisation (a mixed-integer solver over a calibrated hydraulic model) and a topological in-relevance harmonic centrality computed directly on the network's directed graph, weighted by node role (inlet, connection, outfall) and spatial constraints such as slope. The topological method needs no hydraulic simulation and is computationally far cheaper, but the authors caution that using it safely still requires hydraulic competence to validate results – centrality is a fast screen, not a substitute for hydraulic judgement. Their review situates this trade-off within a wider body of prior work spanning heuristic and evolutionary placement (Banik et al.; Tinelli et al.; Yazdi), Bayesian and backtracking approaches (Sambito et al.; Chachula et al.; Guadagno et al.), and other centrality-based methods (Rodríguez-Alarcón & Lozano; Zuluaga et al.; García-Usuga et al.), including a finding from stream-gauge placement that betweenness centrality is particularly effective at identifying bridge points between sub-networks (Halverson & Fleming, 2015).

Ding et al. (2025) take a data-driven sparse-sensing (DSS) approach to a related but distinct objective: reconstructing full-network flow states from a minimal sensor subset, using singular value decomposition to find a reduced basis and QR factorisation with column pivoting to select the most representative physical locations. Applied to a 77-node storm-drainage catchment using an EPA-SWMM-generated training set, reconstruction accuracy (Nash-Sutcliffe efficiency, NSE) rose from a mean of 0.896 with one sensor to 0.993 with ten, and just three sensors achieved a mean observed-event NSE of 0.949. The method was robust to realistic sensor noise (median NSE ≈ 0.981 under 5–15% multiplicative noise) but not uniformly robust to sensor loss: single-location dropout testing found most locations could be lost with a small NSE penalty (0.007–0.038), while two of six tested locations caused much larger losses (0.299 and 0.367) – showing that even a well-optimised sparse layout can depend disproportionately on a small number of irreplaceable locations.

Synthesis

Together, these studies indicate that a defensible sensor-placement methodology needs to combine a risk signal (Malek Mohammadi et al., 2020; Ma et al., 2025; Lundberg et al., 2026) with an observability signal (Simone et al., 2023; Ding et al., 2025), since neither alone identifies where a sensor should go: a high-risk location is not necessarily informative for network-wide state reconstruction, and an information-maximising location is not necessarily where failure consequence is highest. A further consistent theme is that reported factor thresholds, model accuracies, and centrality rankings are specific to the network, climate, and data regime each study used, and require local recalibration and validation – typically against a hydraulic model or incident history – before being applied to a new network. Ma et al. (2026) and Mohammadi et al. (2020) further identify a recurring data-availability asymmetry (physical attributes abundant, hydraulic/environmental/incident attributes scarce) that constrains which methods a given utility can practically deploy without additional data collection.

References

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